



Tutor Jolly School

Name: _____

Grade: _____

Class: _____

Class No: _____

HEAT AND ENERGY: HEAT TRANSFER AND SPECIFIC HEAT CAPACITY

Section A: Multiple Choice Questions

Answer all questions. Choose the correct option for each question.

1. Which of the following mechanisms is primarily responsible for the conduction of heat in metals?

[1Pts]

A. Vibration of molecules only

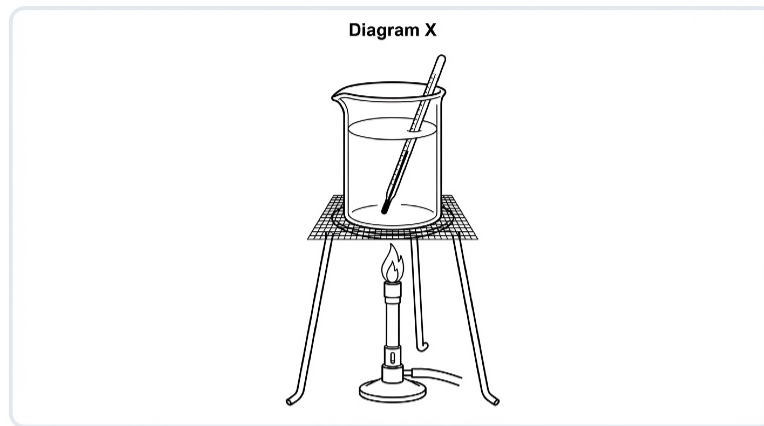
B. Movement of free electrons and lattice vibration

C. Expansion of space between atoms

D. Convection currents within the metal

2. The diagram shows water being heated in a beaker. Which set of arrows correctly represents the convection currents formed in the water?

[1Pts]



- A. Arrows rising from the center and sinking at the sides
- B. Arrows rising at the sides and sinking in the center
- C. Arrows moving horizontally only
- D. Arrows moving randomly in all directions

3. Which of the following surfaces is the best absorber of infrared radiation?

[1Pts]

A. Shiny white surface

B. Dull white surface

C. Shiny black surface

D. Dull black surface

4. Why is the space between the double walls of a vacuum flask evacuated (made into a vacuum)?

[1Pts]

A. To prevent heat loss by conduction and convection

B. To prevent heat loss by radiation only

C. To prevent heat loss by conduction only

D. To keep the flask lightweight

5. A 2 kg block of copper requires 7,800 J of energy to raise its temperature by 10°C. What is the specific heat capacity of copper?

[1Pts]

A. 390 J/(kg·°C)

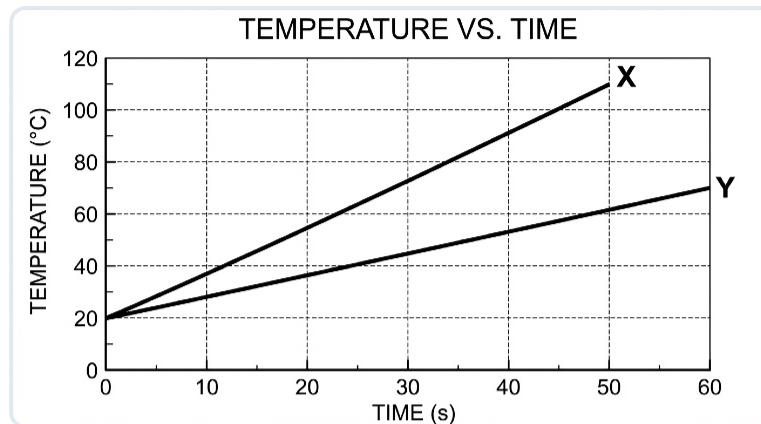
B. 780 J/(kg·°C)

C. 1560 J/(kg·°C)

D. 3900 J/(kg·°C)

6. Two blocks, X and Y, of the same mass are heated by identical heaters. The graph shows the temperature change with time. Which statement is correct?

[1Pts]



- A. Block X has a higher specific heat capacity than Block Y.
- B. Block Y has a higher specific heat capacity than Block X.
- C. Both blocks have the same specific heat capacity.
- D. Block X absorbs more heat energy than Block Y in the same time.

7. Water is used as a coolant in car engines primarily because:

[1Pts]

- A. It is a good conductor of heat.
- B. It has a high specific heat capacity.
- C. It has a low boiling point.
- D. It expands significantly when heated.

8. On a cold winter day, a metal door handle feels colder than the wooden door itself. What is the reason for this?

[1Pts]

A. The metal handle is at a lower temperature than the wood.

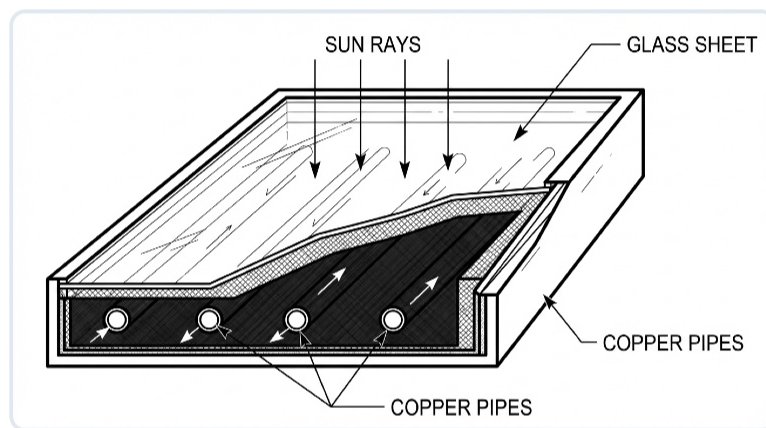
B. Metal has a higher specific heat capacity than wood.

C. Metal is a better thermal conductor than wood.

D. Wood radiates more heat than metal.

9. The diagram shows a solar water heater panel. Why is the pipe inside the panel painted black?

[1Pts]



A. To reflect heat radiation from the sun.

B. To conduct heat better to the water.

C. To absorb maximum heat radiation from the sun.

D. To prevent the pipe from rusting.

10. Which equation represents the principle of conservation of energy when a hot substance is mixed with a cold substance (assuming no heat loss to surroundings)?

[1Pts]

A. Heat gained by cold substance = Heat lost by hot substance

B. Mass of cold substance = Mass of hot substance

C. Temperature rise of cold substance = Temperature drop of hot substance

D. Specific heat of cold substance = Specific heat of hot substance

Section B: Structured Questions

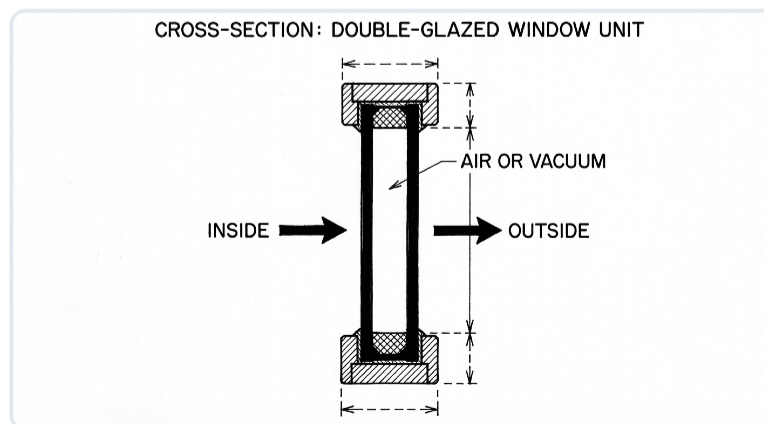
Answer all questions in the spaces provided. Show your working for calculations.

11. A house is fitted with double-glazed windows to reduce energy costs.

[5 Pts]

(a) Describe the structure of a double-glazed window. (2 points)

(b) Explain, in terms of heat transfer mechanisms, how double-glazed windows reduce heat loss from the house during winter. (3 points)

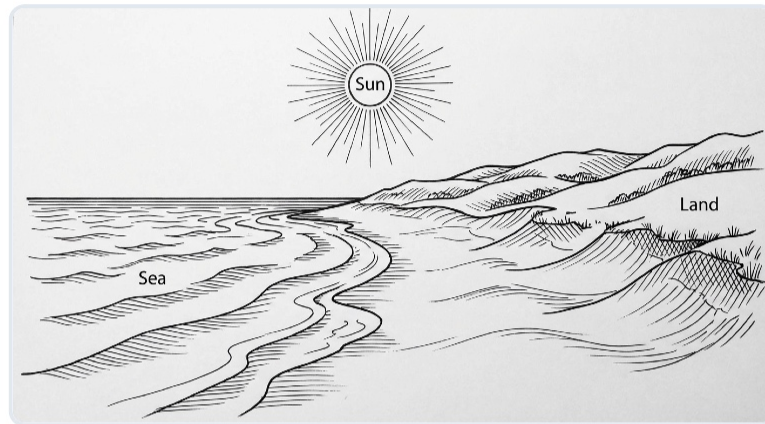


12. The diagram shows a coastal region during the day.

[6 Pts]

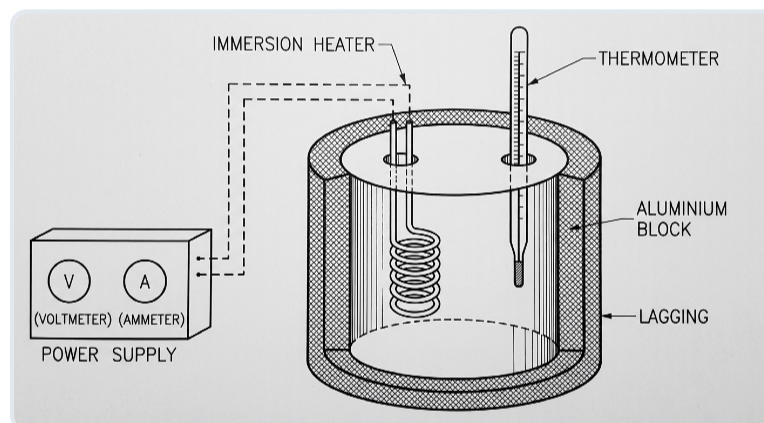
(a) Explain why the land heats up faster than the sea during the day. (2 points)

(b) Describe how this temperature difference creates a 'sea breeze'. Include the movement of air and density changes in your explanation. (4 points)



- 13.** A student performs an experiment to determine the specific heat capacity of an aluminium block using an immersion heater.

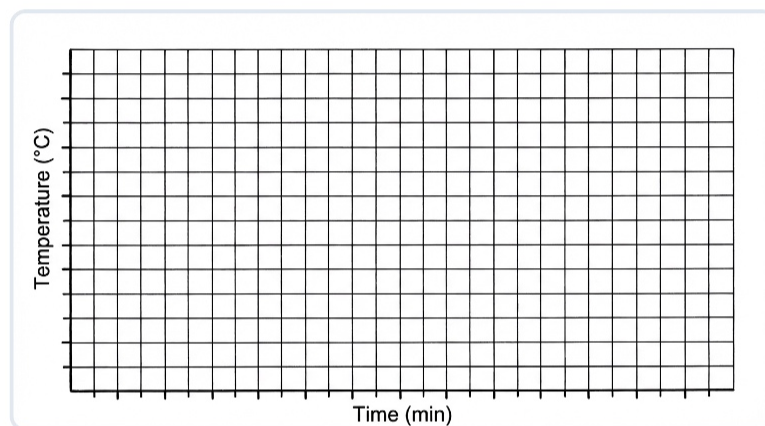
- (a) The heater is rated at 12 V, 2 A. Calculate the power of the heater. (2 points)
- (b) The heater is switched on for 5 minutes (300 seconds). Calculate the total electrical energy supplied. (2 points)
- (c) The mass of the block is 1 kg. The initial temperature is 20°C and the final temperature is 26.5°C. Calculate the specific heat capacity of the aluminium block. (3 points)
- (d) Why is the calculated value likely to be higher than the actual specific heat capacity of aluminium? (1 point)



- 14.** A hot liquid is allowed to cool in a laboratory. The temperature is recorded every minute.

[8 Pts]

- (a) Plot the shape of the cooling curve you would expect on the provided axes (Temperature vs Time). (2 points)
- (b) Explain why the rate of cooling decreases as time passes. (3 points)
- (c) If the beaker containing the liquid were covered with a lid, how would the cooling curve change? Explain your answer. (3 points)



15. The diagram shows a vacuum flask used to keep liquids hot.

[8 Pts]

(a) Identify the feature labeled 'X' (the silvered surfaces) and explain how it minimizes heat loss. (3 points)

(b) Identify the feature labeled 'Y' (the stopper) and explain how it minimizes heat loss. (3 points)

(c) Why is the vacuum between the walls unable to stop heat transfer by radiation? (2 points)

